Goal 2 — Population-scale chimpanzee isoform diversity (NilabjaBh)

Methods

Study objective and analysis design

The aim of Goal 2 was to quantify how frequently reference chimpanzee transcripts are represented across a population-scale collection of long-read RNA-sequencing samples and to summarize their expression support. The final analysis used the organizer-provided compact chimpanzee Isopedia index rather than reconstructing an index from raw reads and BAM files. Consequently, BAM sorting and indexing, per-BAM isopedia profile generation, profile merging, and de novo isopedia index construction were not repeated. The prebuilt index was queried directly with the Group 1 Ensembl transcript annotation. All workflow stages were implemented both as executable notebooks in Goal_2 and as production Python scripts in Scripts; the serial workflow and calculation rules were recorded in master.md.

Input datasets

The population evidence source was the organizer-provided chimpanzee_index.tar.gz archive distributed through SharePoint. The archive contained a precomputed Isopedia index for long-read transcriptomic data. The query annotation was the Ensembl Pan troglodytes geneset release 2025_05 on assembly GCA_028858775.3, supplied by Group 1 as Ensembl_reference_transcripts.gtf.gz. The Ensembl GTF defined the reference transcripts being evaluated; it did not define the population denominator. Analyses were performed with Isopedia version 1.6.6 and Python using pandas, NumPy, and Matplotlib.

Official index acquisition and integrity checks

The anonymous SharePoint folder was opened programmatically and searched for exactly one file named chimpanzee_index.tar.gz. Downloads were resumable and were accepted only when the local byte count matched the remote object. The verified archive contained 6,801,051,558 bytes and. Archive extraction was protected against absolute paths, parent-directory traversal, symbolic links, and hard links. After extraction, the index root was identified from the simultaneous presence of meta.txt, chrom.map, and bptree_*.idx files.

Exactly one valid index root containing 30 files was required. The share URL, byte counts, checksum, extraction time, index path, and index-file count were recorded in index_download_provenance.tsv.

Preparation and validation of the Ensembl query annotation

The compressed Ensembl GTF was read without changing genomic coordinates. Transcript and exon records were grouped by transcript_id. For each transcript, the transcript record was written before its exons, exons were ordered by genomic coordinate, and transcript models were ordered by chromosome, start, end, and transcript identifier. This parent-first derivative was required for reliable Isopedia querying and was distinct from building or sorting the population index. The prepared GTF contained 914,843 feature rows: 102,140 transcript models and 812,703 exon records. Every retained transcript had at least one exon, and no orphan or duplicate transcript model was accepted.

Annotation of reference transcripts with Isopedia

The prepared Ensembl transcript models were queried against the extracted index using:

Data/isopedia isoform -g Data/group1/Ensembl_reference_transcripts.sorted.gtf -i Data/official_57_sample_index/extracted/chimpanzee_index -o Output_files/official_57_sample/annotated.gtf -n 16 --info

The command loaded all 102,140 query transcripts. Isopedia was run with its recorded version 1.6.6 settings, including a 10-bp terminal tolerance/flank setting, minimum read support of one, expectation-maximization maximum of 100 iterations, convergence threshold of 0.01, damping factor of 0.3, and minimum EM abundance of 0.0001. The output was materialized as annotated.gtf.gz. Despite its filename, this file was treated as Isopedia's native transcript-by-sample table rather than as an ordinary nine-column GTF. The native result contained transcript coordinates, identifiers, ranking information, cohort detection fields, and one colon-delimited expression field per indexed sample.

Definition of the 57-sample RNA-seq cohort

The delivered meta.txt contained 60 nonempty lines. Isopedia interpreted the first line as a header and emitted 59 native sample-expression columns. Two columns did not belong to the intended long-read RNA-seq cohort: SRR6713217 was chimpanzee ICE data with library strategy OTHER, and SRR6713218 was orangutan (Pongo abelii) ICE data with library strategy OTHER. Both columns were explicitly excluded. All population and expression statistics were then recalculated from the remaining 57 RNA-seq columns. The native 59-column annotation was retained unchanged as provenance, while every downstream table and figure used a denominator of 57.

Parsing transcript-level evidence

Each retained sample field followed the structure CPM:COUNT:FSM_CPM:FSM_COUNT:EM_CPM:EM_COUNT, with an optional trailing information component. CPM was the sample-normalized transcript abundance, COUNT was the transcript-associated read count, FSM_COUNT represented full-splice-match evidence, and EM_COUNT represented expectation-maximization-assigned support. Parsing was performed in chunks of 5,000 transcripts to avoid holding the complete 57-column expression matrix in memory.

For transcript t in sample s, presence was defined as:

$$\text{Presence}(t,s) = 1 \text{ if } \text{COUNT}(t,s) > 0 \text{ or } \text{FSM\_COUNT}(t,s) > 0 \text{ or } \text{EM\_COUNT}(t,s) > 0; \text{ otherwise } 0.$$

The number of positive samples was:

$$\text{PositiveSamples}(t) = \text{sum over all retained samples of Presence}(t,s).$$

Population frequency was calculated as:

$$\text{PopulationFrequency}(t) = \text{PositiveSamples}(t) / 57.$$

FSM and EM population frequencies were calculated independently as the numbers of samples with FSM_COUNT > 0 and EM_COUNT > 0, respectively, divided by 57. Positive-sample counts were required to lie between 0 and 57.

Expression and read-support summaries

Mean CPM was calculated over all 57 retained samples, including zeros:

$$\text{MeanCPM}(t) = [\text{sum CPM}(t,s)] / 57.$$

Median CPM and maximum CPM were calculated across the same 57-sample vector. Mean-positive CPM was calculated only across samples with CPM > 0. Total read support, total FSM support, and total EM support were calculated by summing COUNT, FSM_COUNT, and EM_COUNT, respectively, across the 57 samples.

A sample_expression_summary.tsv table was produced with one row per retained sample and the following columns: sample, detected_transcripts, summed_cpm, total_read_count, total_fsm_count, and total_em_count. This official sample-expression table—not the removed 17-sample SRA metadata table—was used for all sample-level expression panels. It contained exactly 57 unique sample accessions and was required to agree exactly with the retained annotation columns.

Frequency classes

Transcripts were assigned to mutually exclusive frequency classes using their positive-sample count and population frequency: Not detected (0 samples), Singleton (1 sample), Rare (>1 sample but <10%), Low (10–<25%), Intermediate (25–<50%), Common (50–90%), and Core (>90%). These classes were used to summarize the population distribution and to select representative transcripts for sample-level visualization.

Representative transcript selection

For the transcript-by-sample presence heat map, transcripts were stratified into five prevalence bands: singleton; nonsingleton below 10%; 10–<25%; 25–<50%; and at least 50%. Within each band, up to eight transcripts with the greatest total read support were selected. This produced 40 representative transcripts. Their presence or absence was displayed across all 57 samples.

Discovery saturation analysis

Transcript discovery saturation was assessed using 300 deterministic random permutations of sample order with random seed 20260827. For each permutation, samples were added sequentially and the cumulative union of detected transcripts was recorded after each added sample. At every cohort size k, the mean number of discovered transcripts and the 2.5th and 97.5th percentiles across permutations were reported. The final value was required to equal the observed union of transcripts across all 57 samples, and the cumulative discovery curve was required to be monotonic.

Quality control and validation

The workflow required a nonempty, byte-verified index archive; exactly one extracted index root; valid index marker files; a readable Ensembl GTF; 102,140 query transcripts with exon support; a native Isopedia header; 59 native expression columns; successful removal of the two predefined ICE columns; exactly 57 retained RNA-seq samples; unique sample names; and matching sample sets between the annotation and sample_expression_summary.tsv. Every colon-delimited sample field was required to contain at least six numeric components. Output tables were required to be nonempty, population frequencies were constrained to [0,1], and positive-sample counts were constrained to [0,57]. The workflow stopped at the first failed validation and recorded commands, timestamps, durations, logs, and completion status in official_execution.tsv.

Visualization

Figures were generated with Matplotlib as 300-dpi PNG files and vector PDF files. Figure 2 reported the validated cohort size. Figure 3 summarized cohort-wide FSM and EM expression evidence. Figure 4 displayed per-sample transcript-assigned read support on a logarithmic scale. Figures 7 and 8 summarized continuous population prevalence and categorical frequency classes. Figure 9 compared population support with total read support. Figure 10 displayed the representative transcript-by-sample presence matrix. Figure 12 reported detected Ensembl transcripts per sample. Figure 13 displayed the permutation-based discovery saturation curve.

The compact index did not contain raw FASTQ sequences, per-read lengths, or the pre-index total and mapped-read denominators required for alignment-rate calculations. Accordingly, Figure 5 and Figure 6 were explicitly marked unavailable instead of substituting measurements from a different cohort. The official sample_expression_summary.tsv did not contain tissue or sequencing-platform labels, so tissue-stratified prevalence in Figure 11 was also marked unavailable rather than inferred.

Reproducibility and software organization

The analysis was organized as four executable notebooks: 1_download.ipynb, 2_annotate_group1_ensembl.ipynb, 3_summerise_annotation.ipynb, and 4_generate_fig.ipynb. Equivalent production scripts were numbered 1 through 5 for index acquisition, annotation, summarization, figure generation, and serial execution. The serial workflow can reuse the verified local index without contacting SharePoint by using the --reuse-downloaded-index option. Principal outputs included annotated.gtf.gz, transcript_population_expression.tsv.gz, sample_expression_summary.tsv, annotation_summary.tsv, top_supported_transcripts.tsv, figure source tables, execution logs, and Figures 2–13.

Analysis summary

A total of 102,140 Ensembl transcripts were queried. Evidence was detected for 47,575 transcripts (46.58%), and 42,302 transcripts had full-splice-match evidence. Summed transcript-assigned read support across the retained sample-expression table was 13,717,868. The cumulative discovery set at 57 samples contained 47,575 transcripts.

Figures

Figure 2. Official expression cohort size. The organizer’s native annotation contained 59 expression columns; removal of the two non-RNA-seq ICE columns yielded the validated cohort of 57 long-read RNA-seq samples used for every downstream calculation.

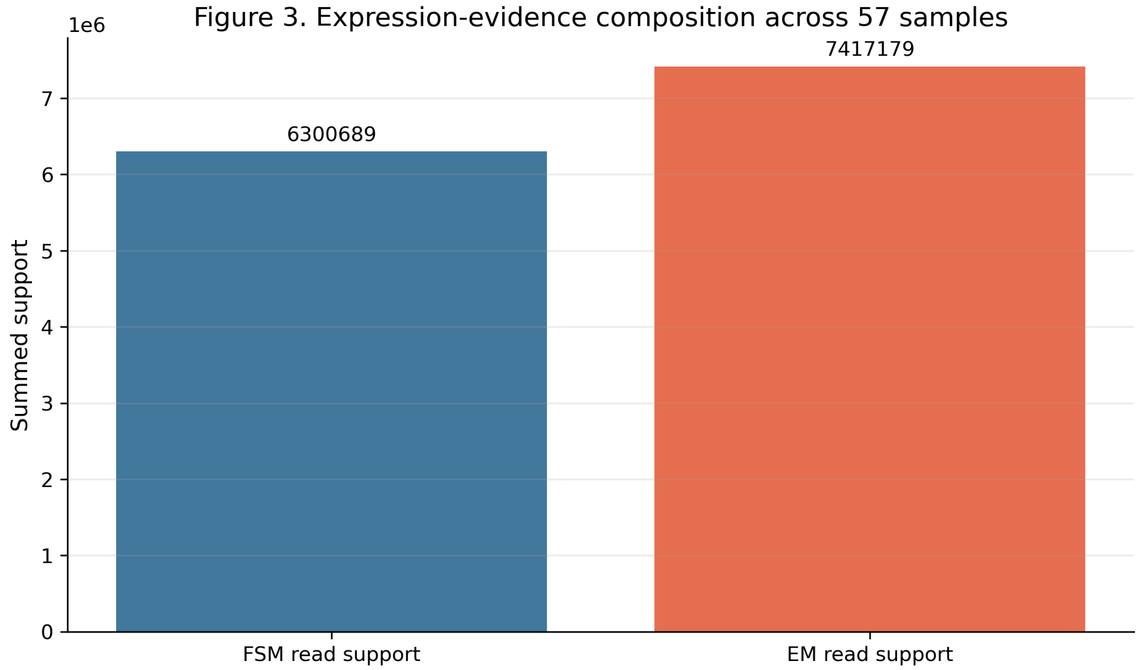

Figure 3. Cohort-wide expression-evidence composition. Bars show the summed FSM_COUNT and EM_COUNT values across the official 57-sample expression summary. These are evidence totals derived from Isopedia’s transcript assignments and are not independent sequencing-platform counts.

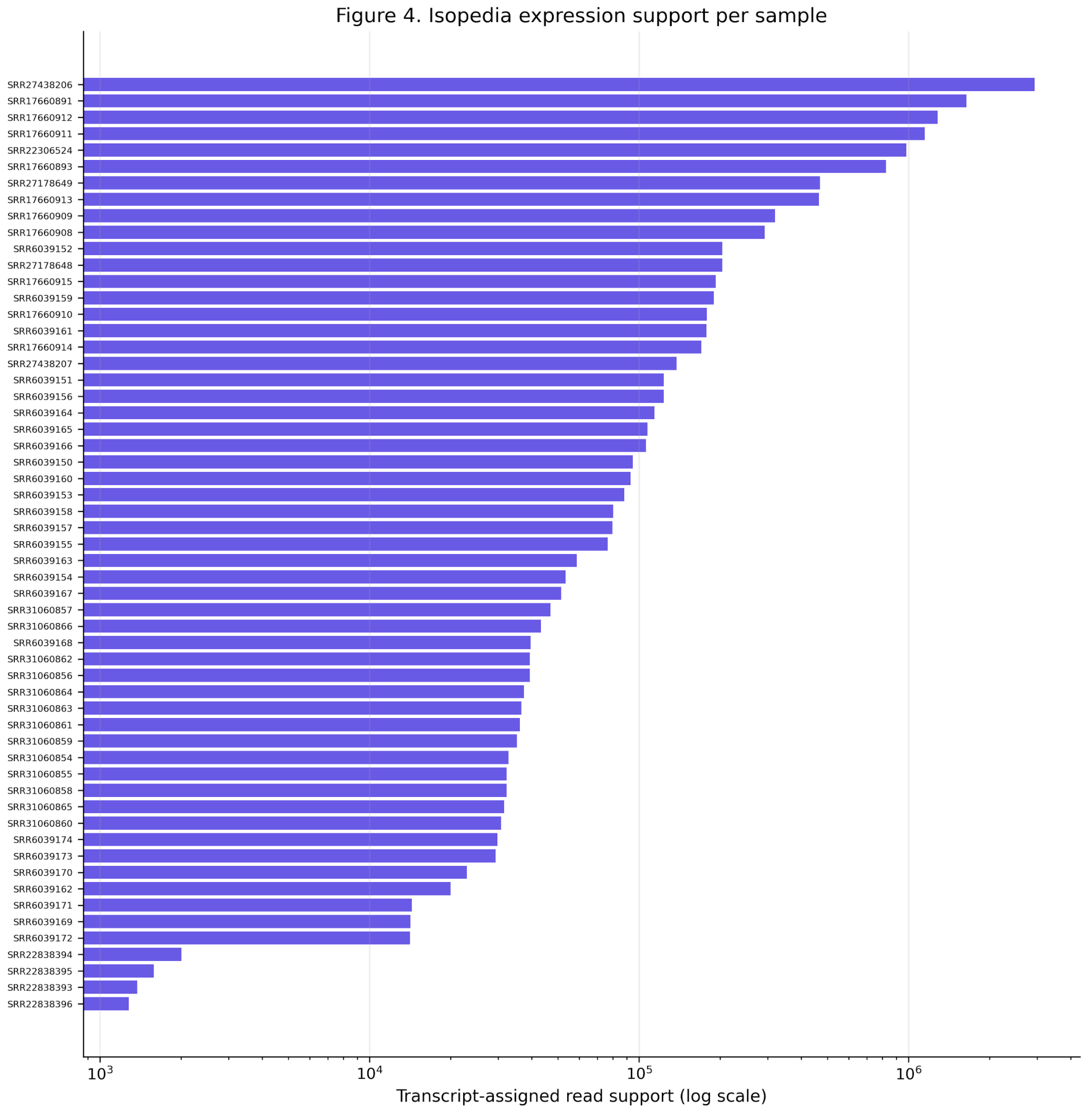

Figure 4. Isopedia expression support per sample. Transcript-assigned read support from sample_expression_summary.tsv is shown for each of the 57 retained samples on a logarithmic horizontal axis, illustrating substantial differences in effective transcriptomic depth.

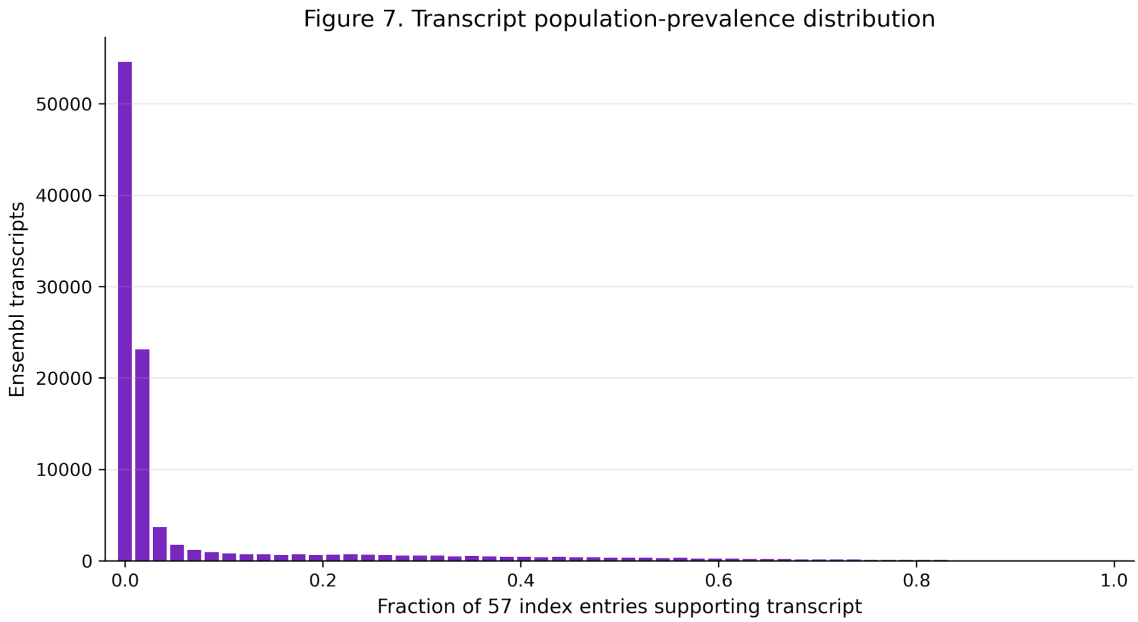

Figure 7. Transcript population-prevalence distribution. For each Ensembl transcript, prevalence was calculated as the number of retained samples with COUNT, FSM_COUNT, or EM_COUNT greater than zero divided by 57. Bars report transcript counts across the full prevalence range from 0 to 1.

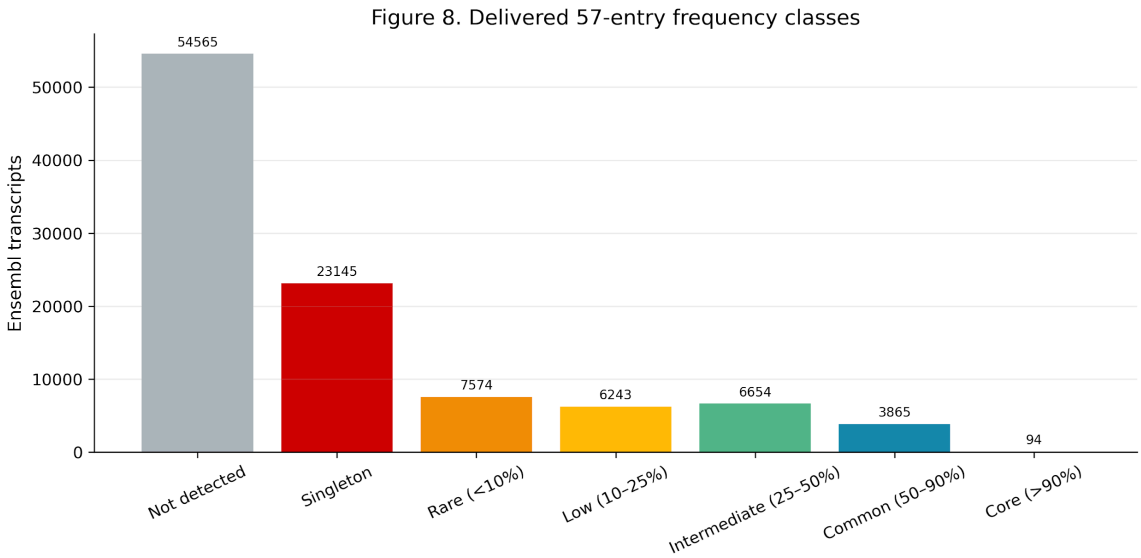

Figure 8. Transcript frequency classes. Ensembl transcripts were classified as not detected, singleton, rare (<10%), low (10–<25%), intermediate (25–<50%), common (50–90%), or core (>90%) using the recalculated 57-sample denominator.

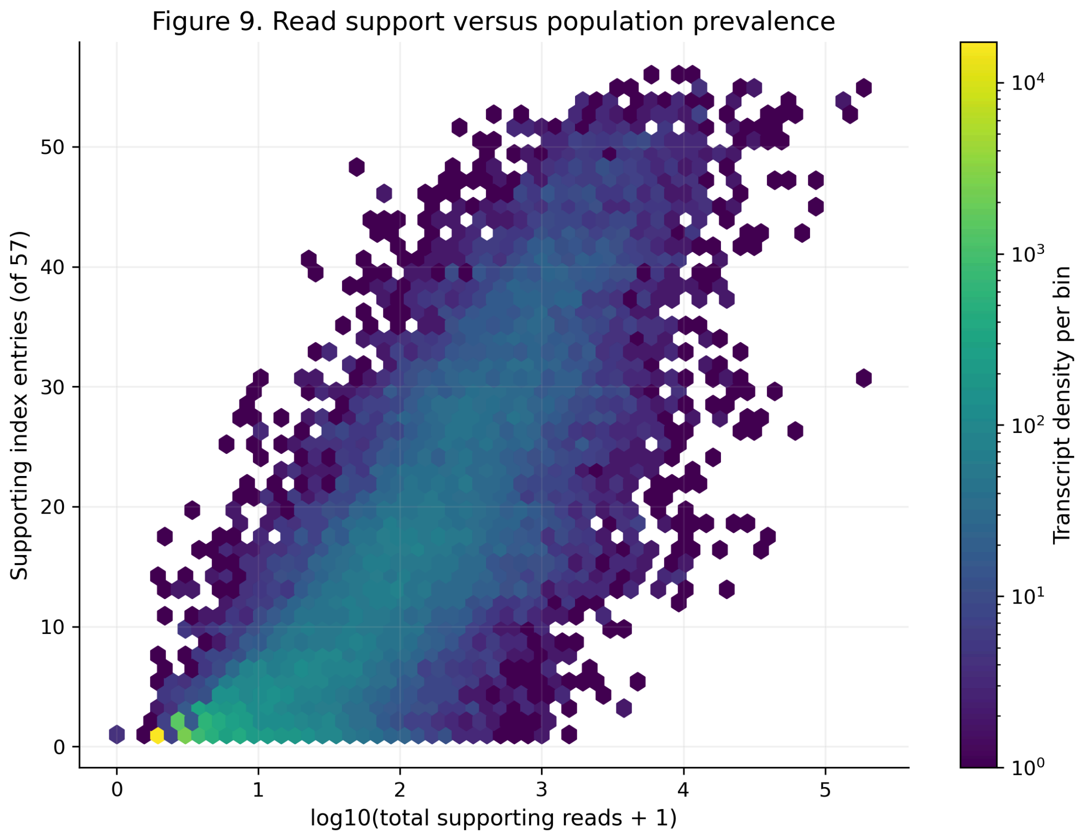

Figure 9. Read support versus population prevalence. Each hexagonal bin summarizes transcripts according to log10(total transcript-associated reads + 1) and the number of supporting samples. Color denotes transcript density, allowing highly supported broadly distributed transcripts to be distinguished from rare or low-support transcripts.

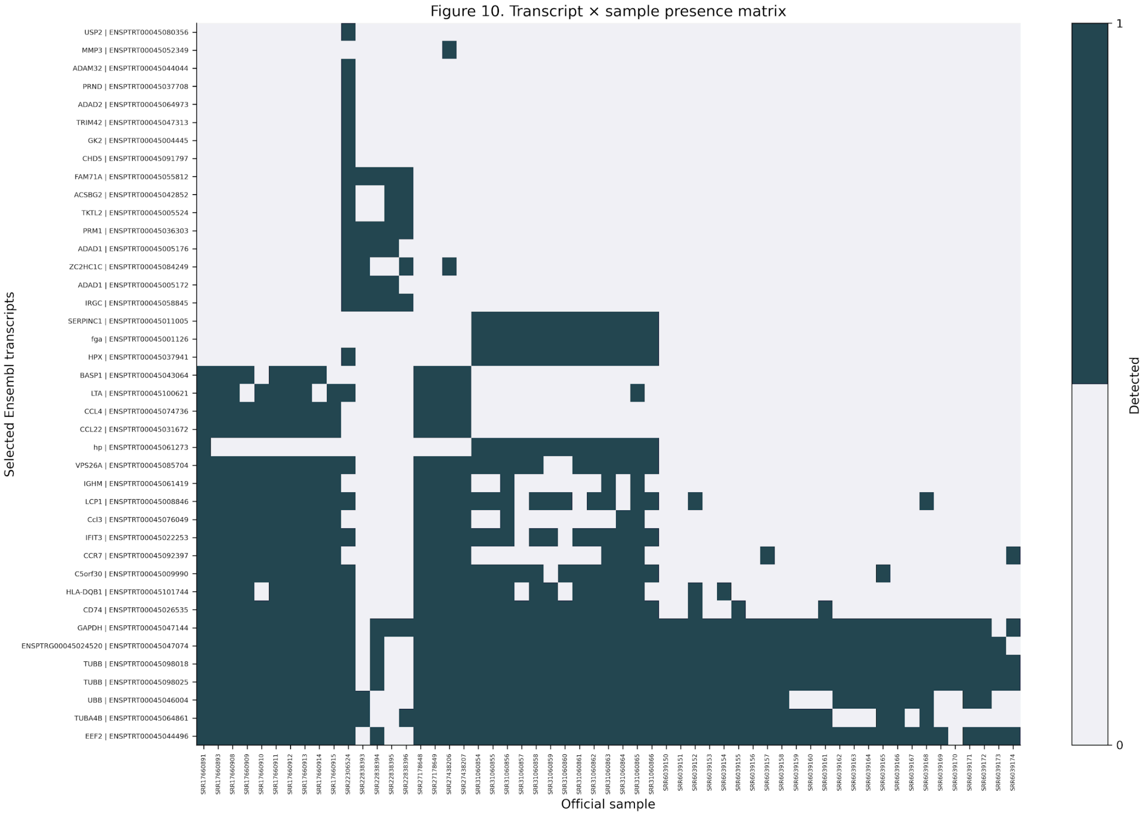

Figure 10. Representative transcript-by-sample presence matrix. Forty transcripts were selected across five prevalence bands, with up to eight high-read-support transcripts per band. Dark cells denote detection by COUNT, FSM_COUNT, or EM_COUNT in the corresponding official sample.

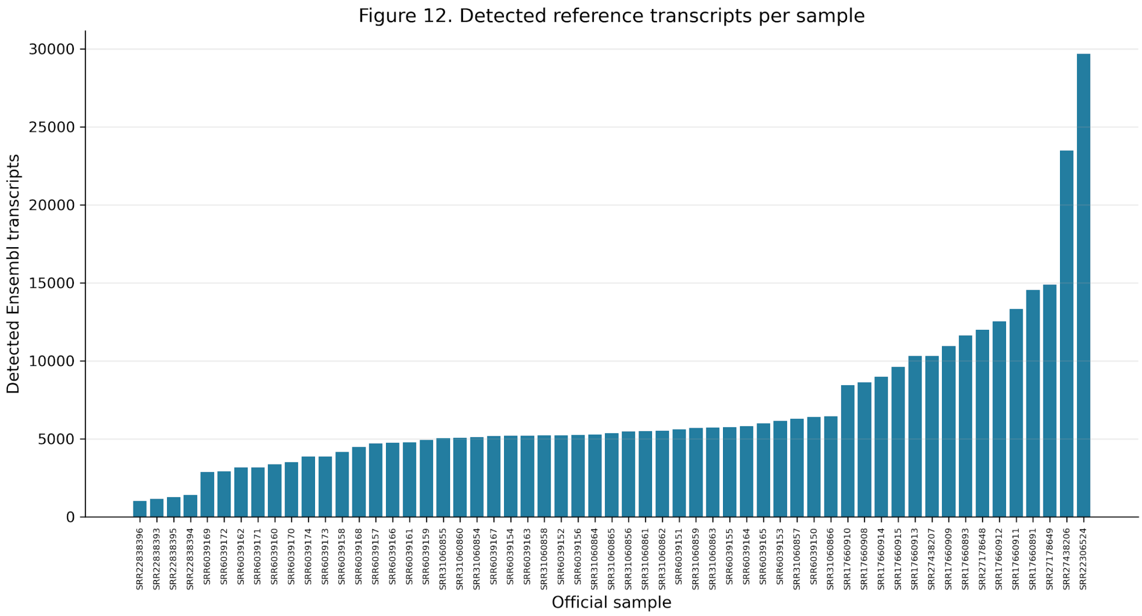

Figure 12. Detected reference transcripts per sample. Bars show the number of Ensembl transcripts with any Isopedia evidence in each of the 57 retained samples, using the validated values in sample_expression_summary.tsv.

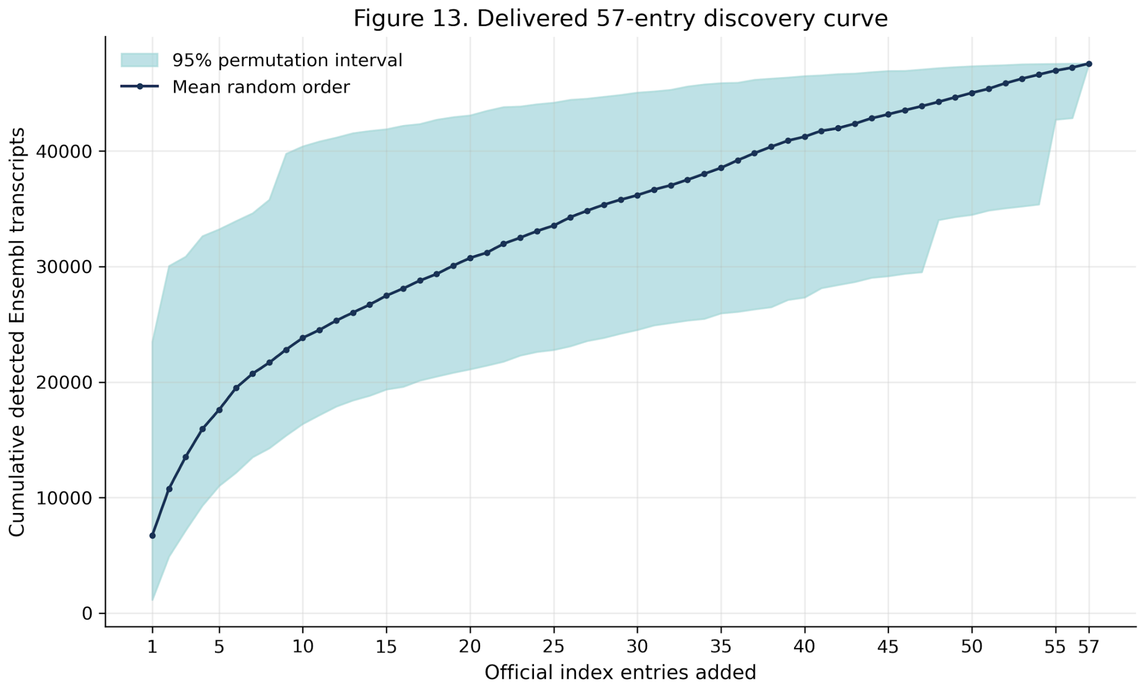

Figure 13. Transcript discovery saturation. Samples were added in 300 deterministic random orders. The line shows the mean cumulative number of detected Ensembl transcripts at each cohort size and the shaded band shows the 2.5th–97.5th percentile interval. The union at all 57 samples contained 47,575 transcripts.